

Akshay Krishna Varma Buddharaju

+1 9188158169 | akvbuddharaju@gmail.com | linkedin.com/in/abuddhar | github.com/AKSHAYKRISHNAVARMA

Summary

AI/ML Developer with 1+ year building and deploying production ML and LLM systems in Python. Built RAG and generative-AI pipelines with PyTorch, Hugging Face, and LangChain, served via FastAPI and deployed on GCP Vertex AI and AWS SageMaker, with Docker, Kubernetes, MLflow, and Airflow across the stack. Delivered measurable results: 85% retrieval-efficiency gain, 50% LLM latency reduction, 73% vulnerability-risk reduction, and 95% NLP classification accuracy.

WORK EXPERIENCE

Machine Learning Engineer, Sports Media INC Feb 2026 – Present

- Built a Python-based MCP server providing developer-facing tooling that automates weekly ad-analytics reporting through campaign-data ingestion, KPI aggregation, and LLM-generated insights.
- Implemented metrics and automation across 15+ ad-performance metrics, reducing manual reporting effort by 70% by eliminating recurring engineering toil.

Machine Learning Engineer, Decentralized Science Lab Aug 2025 – Dec 2025

- Reduced vulnerability risk scores by 73% by building an LLM-based code-analysis and automated-remediation pipeline that analyzed 50 codebases using multi-metric risk re-scoring.
- Built automated data ingestion and preparation pipelines with Apache Airflow, curating a 1,400-sample dataset to support ML training, evaluation, and experimentation.

Machine Learning Engineer, Kennesaw State University Aug 2024 – Apr 2025

- Developed a GNN + BERT Generative AI model in PyTorch for robotic surgical report generation, evaluated at a ROUGE score of 0.81 a 28.6% improvement over baseline.
- Improved report quality from ROUGE 0.81 to 0.83 (+2.47%) by integrating SimGNN with template-based training for stronger structural alignment.

Software Engineer Intern - CoreAI, The Home Depot Inc., May 2025 – Jul 2025

- Designed a video RAG system for Magic Apron and reduced LLM inference latency by 50% through profiling and parallel chunk-based processing of large video inputs, improving responsiveness of retrieval-augmented workflows.
- Improved retrieval efficiency by 85% using optimized embedding generation and similarity-search strategies within a RAG-based knowledge-retrieval system.
- Built chunk-level inference pipelines that processed 100 long-form videos via batch/parallel processing, achieving 100% input coverage for a large, growing customer base.

Machine Learning Engineer, Vidhi Labs Nov 2023 – Dec 2023

- Developed and fine-tuned PyTorch CNN ensembles (ResNet, VGG19, InceptionV3), improving F1-score by 5% for traffic-violation detection under low-latency inference constraints.
- Trained a YOLOv5 object-detection model achieving 92.5% mAP@0.5 for vehicle and sign recognition using optimized computer-vision pipelines.
- Reduced inference latency by 55% (detection) and 45% (classification) through INT8 quantization and structured pruning for compute-efficient deployment.

Software Engineer, INV Technologies Jan 2023 – Oct 2023

- Developed a one-shot face-recognition model (90% accuracy) and deployed it end-to-end on AWS SageMaker with S3-based data storage.
- Built a Seq2Seq neural network for fake-news detection, achieving 95% classification accuracy with sequence-modeling techniques.

SKILLS

Languages: Python, C++, Java, SQL

SDLC & Tooling: Docker, Kubernetes, Git, GitHub Actions (CI/CD), PyTest, Linux, Apache Airflow, MLflow

Cloud & Infrastructure: AWS (SageMaker, Lambda, S3), PySpark, MySQL, MongoDB, Neo4j

ML & AI : Generative AI, LLMs, RAG, Deep Learning, Computer Vision, NLP, Graph Neural Networks, XGBoost, Scikit-learn

Frameworks & Libraries : PyTorch, TensorFlow, Hugging Face, LangChain, FastAPI, Pandas, NumPy, OpenCV, Matplotlib

EDUCATION

Kennesaw State University, Master of Science in Computer Science

Aug 2024 – Dec 2025

Concentrations: Data Science